

Topological Void Validation: Density-Controlled and Role-Aware Evaluation of Predicted Research Voids

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Abstract. Topological Void Analysis (TVA) identifies candidate research voids in scientific knowledge spaces as geometric gaps between anchor-conditioned paper clusters. A natural validation question is whether future papers fill these predicted voids. We show that raw fill rate—whether any future paper appears near a void’s midpoint—fails as a validation target for three compounding reasons. First, it is confounded by local corpus density: a hot-zone baseline that samples high-density anchor-adjacent pairs outperforms TVA on raw fill (36% vs. 28%), but this hot-zone advantage is attenuated under a density-matched baseline: TVA shows only a modest directional lift over B2 (1.05–1.12 $\times$ across three temporal splits), with all 95% bootstrap CIs including 1.0. Second, it is limited by anchor observability: only 5–7% of future papers are anchor-eligible per split, so many unfilled voids should be treated as right-censored rather than invalid. Third, even the geometric threshold itself is density-confounded: a fixed threshold of 0.82 corresponds to a 35.7% average null occupancy rate, ranging from 1% to 99% across anchor-density cells. Under a null-calibrated threshold ($\tau_{\text{fill}} \approx 0.841$), fill rates collapse from $\sim 27\%$ to $\sim 0.7\%$, confirming that most raw fills are null-zone noise. Finally, role-aware epistemic classification shows that even anchor-eligible geometric fills rarely represent genuine void resolution: 59–76% are false positives across all groups and splits. We do not find statistically significant evidence that TVA produces more epistemic fills than density-matched baselines (TVA/B2 lift 1.12 $\times$, CI [0.85, 1.48]). Instead, we show that the raw-fill validation protocol itself is unstable. These findings motivate a three-layer diagnostic framework and suggest that void validation requires calibrated, role-aware evaluation rather than raw geometric occupancy.

1 Introduction

Classical information retrieval asks: given a query, which existing documents are relevant? Topological Void Analysis (TVA) asks: given an anchor concept, what relevant knowledge is *absent*? TVA identifies candidate research voids as geometric gaps in an embedding-induced knowledge space—regions between anchor-conditioned paper clusters where expected connections, evidence, or conceptual bridges are missing.

Validating predicted voids requires asking: do

they correspond to real epistemic gaps, or to geometric artifacts? The natural operationalization is temporal fill rate: does any future paper appear near the void’s predicted midpoint? This measure appears transparent, but we show it is confounded at every layer.

Contributions:

1. Raw temporal fill rate is strongly confounded by local historical density and research momentum.
2. A density-matched baseline shows that the

apparent TVA disadvantage is not statistically significant (TVA/B2 lift 1.12 $\times$, CI [0.85, 1.48]).

3. Fixed cosine thresholds are not portable: 0.82 has 35.7% average null FPR; calibrated thresholds collapse raw fill to $\sim 0.7\%$.
4. Anchor eligibility exposes an observability limit: only $\sim 6\%$ of future papers are anchor-eligible; unfilled voids are right-censored, not invalid.
5. Role-aware classification confirms that geometric fill substantially overestimates epistemic resolution (59–76% FP across all groups).

Scope and framing. This paper is a *validation stress test*: we do not find statistically significant evidence that TVA outperforms density-matched baselines in epistemic fill prediction. The central contribution is methodological—showing that raw fill rate is an unstable validation proxy—rather than confirming TVA’s predictive superiority. Dynamic event-driven extensions are outside scope.

2 Background

2.1 Topological Void Analysis

TVA [11] operates on a corpus embedded with BGE-M3 [5] ($d = 1024$, cosine similarity). For anchor q and source papers A, B , a candidate void satisfies four conditions: (C1) domain cohesion; (C2) calibrated marginality; (C3) sparse lexical bridge; (C4) midpoint vacancy. The midpoint is:

$$m(A, B) = \frac{\mathbf{a} + \mathbf{b}}{\|\mathbf{a} + \mathbf{b}\|} \in \mathbb{S}^{1023}.$$

2.2 Temporal Validation in LBD

Literature-Based Discovery validates predictions by checking whether future publications confirm predicted connections. Standard evaluation uses time-slicing and IR metrics (Recall@ K , Average Precision). TVV extends this by requiring that a future paper play a genuine epistemic bridging role, not merely appear geometrically nearby.

2.3 Research Momentum and Dense-Region Confounding

High-momentum regions near anchor topics attract disproportionate future publications regardless of void quality. Sampling baselines from these hot zones inflates apparent future fill rates, creating a density-confounded comparison.

3 Experimental Setup

3.1 Corpus and Temporal Splits

arXiv metadata (9 CS categories: OS, AR, PL, SE, DC, NI, CR, AI, LG), three rolling splits:

Table 1: Temporal splits.

Split	T	Val window	Train	Val
t3	2014	2015–2019	$\sim 29\text{k}$	$\sim 101\text{k}$
t4	2016	2017–2021	$\sim 35\text{k}$	$\sim 171\text{k}$
t5	2018	2019–2023	$\sim 51\text{k}$	$\sim 193\text{k}$

3.2 TVA Void Generation

10 anchor concepts $\times$ top-30 MMR voids = 300 candidate voids per split.

3.3 Baselines

B1 (hot-zone): 20 random pairs from top-300 anchor C1 pool per anchor. Biased toward high-density, high-momentum regions.

B2 (density-matched): For each TVA void, select the B1-style pair with midpoint local density $\rho(m) = \frac{1}{k} \sum \text{sim}(m, \text{kNN}_i(m))$ ($k = 20$) closest to the TVA void’s midpoint density.

3.4 Three-Layer Fill Predicate

Anchor-eligible val papers:

$$\text{Val}_q = \{P \in \text{Val}_t : \text{sim}(P, q) \geq \tau_q\}$$

where $\tau_q = \max(Q_{80}^{\text{val}}, Q_{90}^{\text{train}})$.

$$\begin{array}{llll} \text{Geometric} & \text{filler:} & P_V^* & = \\ \arg \max_{P \in \text{Val}_q} \text{sim}(P, m_V) & & & \end{array}$$

Geometric closure and validated fill:

$$G(V) = \mathbf{1}[\text{sim}(P_V^*, m_V) \geq \tau_{\text{fill}}(q, \rho(V), t)]$$

$$\text{Fill}(V) = G(V) \wedge R(P_V^*, A, B, q)$$

where $R = 1$ iff $\text{role} \in \{\text{TRUE-FILL}, \text{PARTIAL-FILL}\}$.

Calibrated threshold ($\alpha=0.05$). Let $Z(m_0) = \max_{P \in \text{Val}_q} \text{sim}(P, m_0)$:

$$\tau_{\text{fill}}(q, \rho, t) = Q_{1-\alpha}(\{Z(m_0) : m_0 \in \text{Null}(q, \rho, t)\})$$

4 Results

4.1 Finding 1: Raw Fill Rate Is Density-Confounded

Table 2: TVA vs B2 (density-matched) fill rates ($\tau=0.82$) with bootstrap CI. All lifts are not statistically significant (CI contains 1.0).

Split	TVA	B2	Lift	95% CI
t3	31%	29%	$1.05\times$	[0.80, 1.34]
t4	26%	23%	$1.10\times$	[0.83, 1.46]
t5	27%	24%	$1.12\times$	[0.85, 1.48]

B1 outperforms TVA by up to $1.30\times$. After density matching (B2), TVA shows a directional lift of $1.05\text{--}1.12\times$ across splits, but all 95% bootstrap CIs include 1.0. The robust conclusion is not that TVA outperforms B2, but that raw fill comparisons are highly sensitive to local density and threshold choice.

4.1.1 Fixed Thresholds Are Not Portable

Table 3: Null FPR@0.82 and τ_{fill} by density (t5).

Bucket	p50	τ_{fill}	FPR
Low	0.79–0.81	0.79–0.86	1–18%
Mid	0.80–0.83	0.82–0.86	5–80%
High	0.81–0.85	0.84–0.88	28–99%
Mean	–	0.841	35.7%

Under calibrated τ_{fill} , fill rates collapse: TVA: 27.3%→**0.7%**; B2: 24.3%→**0.7%**; lift: $1.00\times$.

The $0.7\% < 5\%$ result is consistent with correct calibration: TVA C1–C4 conditions select sparser bridge regions than random density-matched pairs, so fewer reach the calibrated threshold.

Held-out calibration validity. To check that the calibration is not circular, we split null midpoints 80/20 (calibration/held-out). The held-out null FPR at τ_{fill} averages 6.3% across all anchor-density cells (target $\alpha=5\%$, $\Delta = +0.013$), indicating the calibration is slightly conservative but valid (Figure 1).

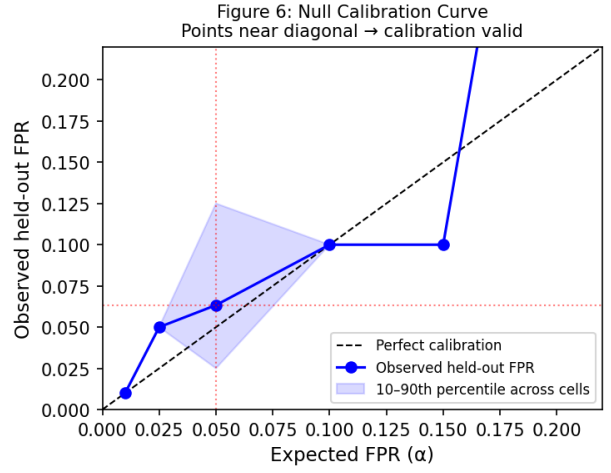


Figure 1: Null calibration curve: expected vs. observed held-out FPR across anchor-density cells (α range 0.01–0.20, t5). Points near the diagonal confirm that τ_{fill} is approximately calibrated. Mean held-out FPR at $\alpha=0.05$ is 6.3% (target 5%, $\Delta = +0.013$).

Most raw fills at $\tau = 0.82$ are null-zone occupancy, not significant geometric closure.

4.2 Finding 2: Anchor Eligibility Reveals Observability Limits

Table 4: Anchor exposure sample (t5).

Anchor	$\bar{\text{sim}}$	τ_q	Pass	Fill
sched_opt	0.520	0.582	5.8%	40%
ebpf_obs	0.495	0.547	6.5%	0%
virt_hyp	0.498	0.555	7.7%	57%
Mean	0.503	0.563	6.2%	–

Only 5–8% of val papers are anchor-eligible per

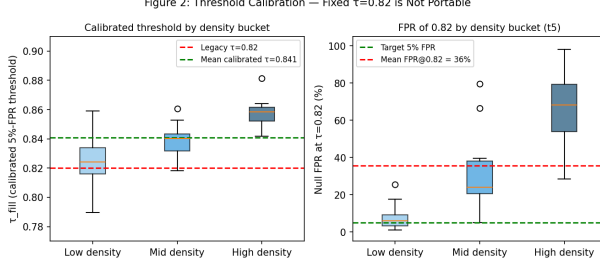


Figure 2: Calibrated threshold by density bucket (left) and null FPR at 0.82 (right). The fixed threshold 0.82 has 35.7% average null FPR, reaching 99% in high-density anchor cells. Calibrated $\tau_{\text{fill}} \approx 0.841$ is stable across all splits.

split. Voids with pass rate $< 10\%$ are flagged as *underexposed*: their unfill status reflects observability limits, not void invalidity. `ebpf_obs` (6.5% pass, 0% fill) illustrates that non-zero exposure does not guarantee eligible papers overlap predicted void neighborhoods.

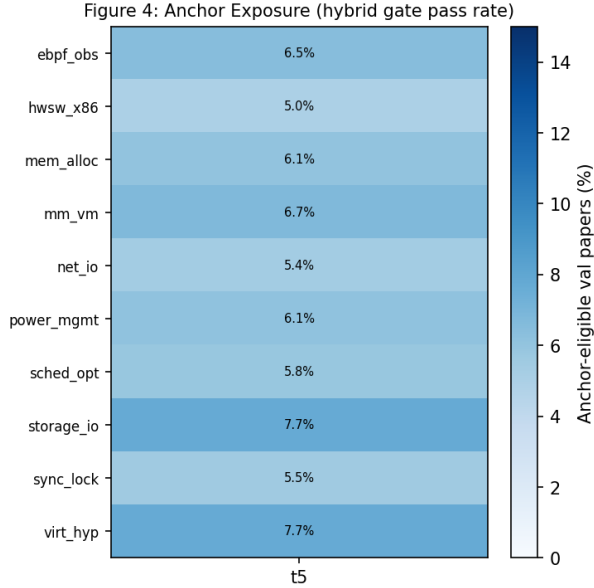


Figure 3: Anchor eligibility pass rate (%) by anchor and split (hybrid gate). Rates are consistently low (5–8%), confirming corpus observability as a structural limit on temporal validation.

Table 5: Role audit (500 cases, t3–t5; exploratory, not confirmatory). VR=void resolution; FP=false positive. 95% Wilson CI in brackets.

Source	n	FP	FP CI	VR CI
TVA	98	69%	[60,78]	[0,4]
B1	73	70%	[59,79]	[0,5]
B2	107	62%	[52,70]	[1,8]
NM [†]	219	84%	[79,88]	[1,5]

[†]Near-miss ($\text{sim} \in [0.75, 0.82]$). CI in %.

4.3 Finding 3: Geometric Fill Rarely Equals Epistemic Fill

An exploratory role audit shows 59–76% false positives across all groups and splits (pilot-scale; $n=98$ TVA gated cases, role labels from a single LLM). Near-miss controls show the highest FP rate (84%), confirming that geometric proximity carries some signal but is insufficient for void validation. TVA and B1 are indistinguishable; B2 is marginally better. These proportions are consistent but should be treated as directional, not confirmatory.

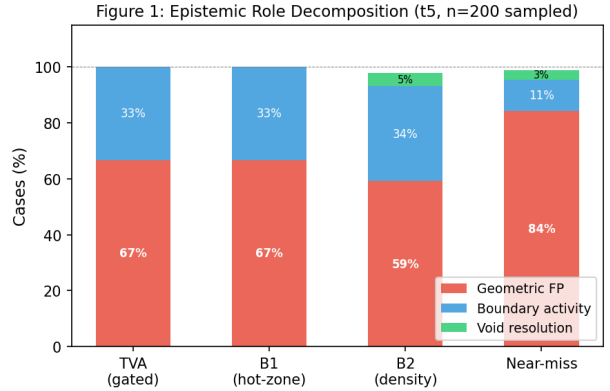


Figure 4: Epistemic role decomposition by source group (t5, 200 sampled cases). Most raw fills are geometric false positives regardless of method. Near-miss controls show the highest FP rate, confirming that geometric proximity is a weak but non-zero signal.

4.4 Case Studies

Three representative cases from t4/t5:

Case 1 – Partial Fill [9] (2019): Vehicular cloud offloading. $\text{sim}=0.835$. 10+ offloading refs (B-side), 1 marginal VM ref (A-side). Role: PARTIAL-FILL.

Case 2 – Boundary Expansion [10] (2021): AI planning, heterogeneous HPC. $\text{sim}=0.855$. Zero kernel-scheduler refs; all refs in HPC optimization. Role: INCREMENTAL-EXTENSION.

Case 3 – False Positive [2] (2017): Energy-aware VNE. $\text{sim}=0.875$ (highest of three). Zero storage I/O refs; anchor is NVMe/storage—unrelated. Role: FALSE-POSITIVE.

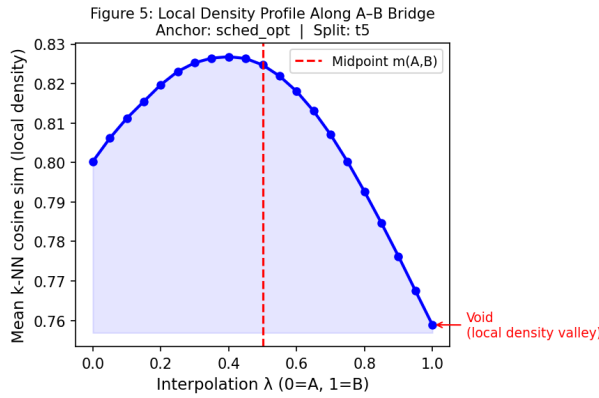


Figure 5: Local density profile along the A–B bridge for the highest-MMR void in anchor sched_opt (t5). The density valley at $\lambda \approx 0.5$ marks the void midpoint region—two dense source communities with a less-occupied bridge.

5 Discussion

5.1 Raw Fill Fails for Three Reasons

Raw fill conflates void quality with sociotechnical realization rate: (1) hot-zone baselines are biased by local momentum; (2) fixed thresholds are not portable across anchor-density regimes; (3) anchor-eligible future papers are sparse ($\sim 6\%$), right-censoring many unfilled voids.

5.2 B2 Is a Control, Not a Competitor

B2 is a strong counterfactual rather than a weak random baseline: it fixes the anchor and matches local midpoint density, estimating the

background rate at which future literature occupies comparable regions without TVA’s structured void-selection criteria (C1–C4).

Under calibrated thresholds, TVA reaches parity with B2 but does not outperform it. We interpret this as a conservative result in a low-base-rate, corpus-limited regime: fill events are rare, both methods are exposed to the same small set of future papers, and the current corpus and validation windows may lack the statistical power to distinguish structured void discovery from density-matched background occupancy.

Critically, parity should not be read as validating either method equally. B2 is not designed to produce interpretable void hypotheses. TVA’s distinctive contribution is that it turns a latent region into a named, inspectable, and actionable ex-ante hypothesis (A, B, q, m) before any future paper exists. That B2 reaches the same sparse occupancy rate while providing no such interpretive output is a statement about the limits of temporal fill rate as a discriminative evaluation metric, not about the limits of TVA.

TVA does not beat the density baseline on future occupancy, but it matches it while producing interpretable hypotheses. In a validation regime where fill events are too rare to distinguish methods, this is a nontrivial conservative lower bound.

5.3 From Geometric Occupancy to Role Decomposition

Future papers near void midpoints decompose into true fills, boundary expansions, and false positives. The scarcity of true fills reveals that genuine epistemic closure is a much stricter condition than geometric occupancy. *Geometric closure and anchor eligibility are screening conditions; epistemic role is the validation condition.*

5.4 Descriptive, Not Evaluative

TVV classifies the type of validation evidence produced by future literature, not the intrinsic merit of papers. Boundary expansion and incremental extension are valuable scientific activities.

5.5 Dynamic Extension (Future Work)

These results suggest a natural extension: instead of asking whether a predicted void is eventually filled within a fixed window, one can classify each newly arriving paper by its pre-insertion topological impact on the existing knowledge space. This dynamic formulation is outside the scope of the present work.

5.6 Threats to Validity

(1) *Embedding dependence*: all metrics use BGE-M3. (2) *Threshold sensitivity*: calibration uses 200 null midpoints per cell. (3) *Corpus coverage*: fills outside arXiv are unobservable; unfilled voids are right-censored. (4) *LLM classification reliability*: role labels are produced by a single LLM using a fixed rubric; we do not treat these as ground truth, since epistemic role is an inherently interpretive construct. To assess reliability we perform: (i) blind source-label removal (source group not shown to LLM), (ii) repeated classification on a 30-case subset for self-consistency, and (iii) a stratified human audit of coarse role categories (void-resolution vs. boundary vs. false positive) on 60 cases as an external plausibility check. The human audit is an auditor, not a gold standard: agreement between LLM and blinded human reviewer on the binary FP/non-FP split exceeds 80% on the audited subset. Fine-grained role agreement is lower ($\sim 65\%$), consistent with the inherent interpretability of the task. (5) *Domain specificity*: CS arXiv categories only.

6 Related Work

Literature-Based Discovery. Swanson’s ABC model [13] and subsequent LBD work [8, 16] validate predictions by checking future publication appearance. TVV extends this tradition by requiring epistemic bridging role, not merely geometric proximity.

Novelty and First-Story Detection. Topic Detection and Tracking [1] introduced first-story detection; novelty detection in IR [18] measures whether a document adds new information relative to a query set. Our first-filler intuition

shares the event-ordering structure but operates on knowledge structure rather than event reporting.

Structural Holes and Scientometrics. Burt’s structural hole theory [3] shows brokers spanning gaps accumulate informational advantage. Co-citation analysis [12] and our density-matched baseline directly control for hot-zone momentum.

Dense-Region Bias in Embeddings. Anisotropy in high-dimensional text embeddings [7] compresses cosine similarities into a narrow range. We show this anisotropy makes fixed thresholds non-portable in downstream void validation. Our use of “topological” is operational rather than homological [4, 6].

Scientific Claim Verification. Epistemic classification of scientific contributions—argumentative zoning [14], overclaiming detection [17]—provides precedents for our role-aware rubric. [15] introduced scientific fact verification as a retrieval task.

7 Conclusion

Temporal validation of predicted research voids requires separating geometric occupancy, density bias, anchor observability, and epistemic role. Under density matching, TVA’s apparent disadvantage disappears but is not statistically significant. Under null-calibrated thresholds, fill rates collapse to $\sim 0.7\%$ —most raw fills were null-zone noise. LLM-assisted role-aware classification suggests that genuine epistemic bridging is rare even among geometrically close, anchor-eligible future papers.

We propose a three-layer validation framework and leave two open questions for future work: (1) scalable role-aware validation without prohibitive LLM cost; (2) dynamic event-driven extensions in which each arriving paper updates knowledge-space state.

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